Chapter 1
Monte Carlo methods for pricing American Options

Raul Chavez Aquino, Fabian Bastin, Maria Benazzouz, and Mohamed Kharrat

Abstract American options are widespread in the financial market. We review various popular techniques used to value American options, as well as Malliavin calculus and recent approaches proposed in machine learning, and examine their performance on synthetic and real data. Our preliminary results confirm that pricing an American put option on a single asset can be efficiently done using regression approaches, and random forests are competitive in terms of accuracy and computation times. Malliavin calculus, despite its interesting mathematical properties, is not competitive for American option pricing, and neural networks are difficult to design in the context of options. Variance reduction, achieved here by means of control variates, is a crucial tool to obtain realiable results at a reasonable cost.

Key words: American options, Monte Carlo, dynamic programming, variance reduction, control variates

Raul Chavez Aquino
Department of Economics, Université de Montréal, Montréal, QC, Canada, e-mail: raul.chavez.aquino@umontreal.ca

Fabian Bastin
Deppartment of Computer Science and Operations Research, Université de Montréal, and CIRRELT, Montréal, QC, Canada, e-mail: bastin@iro.umontreal.ca

Maria Benazzouz
Department of Economics, Université de Montréal, Montréal, QC, Canada, and Desjardins, Laval, QC, Canada, e-mail: maria.benazzouz@desjardins.com

Mohamed Kharrat
Department of Mathematics, Jouf University, Saudi Arabia, and Laboratory of Probability and Statistics LR18ES28, Sfax University, Tunisia, e-mail: mohamed.kharrat@fphm.rnu.tn

1.1 Introduction

Since their official recognition as a financial tool in 1973 with the creation of the Chicago Board Options Exchange, options have attracted a lot of interest by investors, traders and academicians. An option gives the right, but not the obligation, to buy (call option) or sell (put option) an asset at a predetermined price, either at a fixed time T, called the maturity, or at times in a set $\mathcal{T}$, but no later than T. An American option is defined as an option giving its holder the privilege to exercise it at any time during its life. In order to benefit from this privilege, the holder of this type of option must exercise at the best possible time. The option is usually based on underlying price series (stock price, interest rate, index value, etc.) whose random fluctuations are modeled using stochastic processes. The main difficulty with pricing American options is getting a reliable estimate of the continuation value.

One of the first techniques proposed to valuate financial options is the binomial tree, introduced by Cox et al. [16] and Rendleman and Bartter [40]. Binomial trees rely on the discretization of the stochastic differential equation governing the assets value evolution, and do not generalize well to high dimensional portfolios and stochastic processes other than the geometric Brownian motion. An alternative, introduced by Boyle [10] in the context of option pricing, is the Monte Carlo simulation of asset price trajectories. The value of an American option at some time, given a scenario, is expressed as the maximum between the instantaneous exercise price and the expected continuation value, that is the optimal return that can be obtained if one delays the exercise at a later stage. In their seminal paper, Longstaff and Schwartz [33] use dynamic programming and Monte Carlo simulation to estimate the optimal stopping time frontier, relying on least squares regression to approximate the continuation value. Clément et al. [15] analyze the theoretical properties of the method, that can provide lower bounds in expectation, close to the true option prices.

The continuation value can also be estimated with other techniques. Capitalizing on the formulation proposed by Lions and Régnier [32], Bally et al. [3] explore the use of Malliavin calculus [34, 35]. Their work is further explored by Caramellino and Zanette [13], stressing on the importance of an indirect variable, while Kharrat and Bastin [28] develop the expression of Malliavin weights for American options under a stochastic volatility. Bouchard and Warin [9] compare regression approaches and Malliavin calculus, but without variance reduction techniques. The use of Malliavin calculus in financial engineering had recently been covered by Alòs and Lorite [1], but without consideration of options, and we refer to Pascucci [37] for a more specific coverage. Ruf and Wang [41] review the use of neural networks for option pricing, while Rabia [38] considers the use of random forests, reporting promising results.

All these approaches can greatly benefit from the use of variance reduction techniques, although many of the previously cited works do not take full advantage of them. In particular, Rasmussen [39] proposes to use European options, valued at the exercise times, as control variates, and reports significant improvement in the option valuation accuracy when combined with the Longstaff and Schwartz's approach. Dion and L'Ecuyer [17] explore the use of quasi-Monte Carlo draws. Another strategy to reduce the variance is the multilevel Monte Carlo technique,

proposed by Giles [20]. The main idea is to sum estimators built on multiple sets of simulations with different time steps. Such estimators can be obtained using any of the reviewed dynamic programming approaches. Mixed results have however been reported for American options (see for instance Wu [43]), and Belomestny et al. [5] suggest to rather consider levels corresponding to different degrees of approximation of the continuation values. They illustrate the method on a multi-assets option, using the mesh method [12] and the regression approach [33], in combination with the simpler control variate proposed by [12], reporting promising results along with complexity reduction guarantees.

In this chapter, we numerically compare these techniques to valuate an American option in a simple, yet reasonable, assumptions framework, and derive some practical guidelines, emphasizing the importance of variance reduction techniques. Other techniques have been proposed to price American options, in particular the stochastic mesh method [12]. However, regression methods as the one proposed by Longstaff and Schwartz, when combined to variance reduction techniques, have been shown to outperform the stochastic mesh method, in accuracy and computational time [42]. We will therefore not consider such approaches.

The rest of this chapter is organized as follows. We introduce the American option pricing problem in Section 1.2, and present the binomial tree method in Section 1.3. Section 1.4 covers dynamic programming approaches and we discuss the use of control variates in Section 1.5. Numerical comparisons are performed in Section 1.6. We conclude and present future research avenues in Section 1.7.

1.2 American option pricing

The problem of pricing, or valuing, an American option consists of finding an optimal exercise strategy and valuing the expected discounted payoff from this strategy. Given the underlying asset value S_t, let denote by V_t the option value at time t, defined as

$$\beta_t V_t(S_t) = \sup_{\tau \in \mathcal{T}(t,T)} \mathbb{E}_t[\beta_\tau X_\tau(S_\tau) \mid S_t], \tag{1.1}$$

where $\{X_t\}_{0 \le t \le T}$ is the payoff process with the discount factors β_t, $t \in [0,T]$, with $\beta_0 = 1$, and $\mathcal{T}(t,T)$ denotes the class of stopping times satisfying $t \le \tau \le T$. The American option value is defined as V_0, i.e. the value at time 0.

We assume that the factors β_t, $t \in [0,T]$ are deterministic and the asset value process $\{S_t\}_{0 \le t \le T}$ is Markovian and exogenous, i.e. is not affected by the decision to exercise the option or to wait. For simplicity, we only consider an option on a single asset without dividends. The exercise price at time t of a put option is

$$X_t(S_t) = (K - S_t)^+ := \max\{K - S_t, 0\},$$

where K is the strike. For a call option, the exercise price is $\max(S_t - K)^+$. If the discount factors β_t, $t > 0$, are less than one, it can be shown that the optimal

exercise of an American call option is at the expiration date T [36]. The option is then equivalent to an European call option with maturity date T, whose value at time t is

$$\beta_t V_t^E(S_t) = \beta_T \mathbb{E}[X_T(S_T) \mid S_t],$$

and we will focus on put options. We can similarly define an European put option by allowing to exercise the option at time T only. Obviously, $V_0^E(S_0) \leq V_0(S_0)$.

Solving (1.1) is usually intractable, and the problem is simplified by restricting the set of possible exercise times. An usual approximation is obtained with the Bermudan option, allowing the option holder to exercise it at equidistant times $t_m = m\Delta t$, for $m = 0, \ldots, M$, and $\Delta t = T/M$. Denoting $\mathcal{T}_B(0,T)$ the class of stopping times satisfying $\tau = m\Delta t$, $m \in \{0, \ldots, M\}$, the value of the Bermudan option is

$$V_0^B(S_0) = \sup_{\tau \in \mathcal{T}_B(0,T)} \mathbb{E}[\beta_\tau X_\tau(S_\tau) \mid S_0] \leq V_0(S_0).$$

Under mild conditions, $V_0^B(S_0)$ converges to $V_0(S_0)$ when m grows to infinity.

It is also often important to measure the sensitivity of the option price with respect to some initial conditions. The most common is the Delta, that measures the change in the option price as a result of a change in the value S_0 of the underlying. The other options valuation variables being constant:

$$\Delta(S_0) = \frac{\partial}{\partial S_0} V_0(S_0),$$

where we have stressed the dependency of the option value towards the initial asset value. All the techniques covered in this chapter can be adapted to compute the Delta, that can be approximated by finite difference, using common random numbers for the simulation-based methods covered in Section 1.4. While we have computed them during our numerical experiments, we do not report their value for conciseness.

In the simplest case, the asset value S_t is often assumed to follow the Black-Scholes model, that can be described with the partial differential equation

$$dS_t = rS_t dt + \sigma S_t dB_t, \tag{1.2}$$

where r is the risk-free interest rate, σ is the volatility, and B_t is a standard Brownian motion, and a known initial asset value $S_0 > 0$. The asset price at time t can also be expressed as

$$S_t = S_0 \exp(ht + \sigma B_t)$$

where $h = r - \frac{1}{2}\sigma^2$. Under the Black-Scholes model, it is possible to derive analytical expressions of the price of simple options, as the European option (see for instance Higham [23, Chapter 8]). Moreover, we have

$$\beta_t = e^{-rt}, \quad \beta_{\Delta t_i} := \beta_{\Delta t} = e^{-r\Delta t}, \text{ for } i = 0, \ldots, M-1.$$

Given the time discretization $\{t_i = i\Delta t,\ i = 0, \ldots, M\}$, we define the continuation value at time t_i as the discounted conditional expectation $\beta_{\Delta t_i} \mathbb{E}[V_{i+1}(S_{i+1}) \mid S_t]$,

where $V_i(S_i)$ is defined as in (1.1). As the asset value process is Markovian and exogenous, if the option has not yet been exercised the option value at time t_i is

$$V_{t_i}(S_{t_i}) = \max\left\{X_{t_i}(S_{t_i}), \beta_{\Delta t_i}\mathbb{E}\left[V_{t_{i+1}}(S_{t_{i+1}}) \mid S_{t_i}\right]\right\}. \tag{1.3}$$

The rationale is to exercise if the exercise price surpasses the continuation value. However, the continuation value has no analytical expression and must be approximated. We now review several approaches and compare them on simple examples.

1.3 Binomial tree method

A simple approach to price a single-asset option is the binomial model, introduced by Cox et al. [16]. At each time step, we assume that the asset value can either go up by a factor u, with a probability p, or down by a factor d, with a probability $1-p$. This allows to represent M scenarios with a recombined tree, as in Figure 1.1. If the asset value follows the Black-Scholes model (1.2), the discretization of the process on the binomial tree with $p = \frac{1}{2}$ leads to set

$$u = exp\left(\sigma\sqrt{\Delta t} + \left(r - \frac{\sigma^2}{2}\right)\Delta t\right), \quad d = exp\left(-\sigma\sqrt{\Delta t} + \left(r - \frac{\sigma^2}{2}\right)\Delta t\right)$$

(see for instance Higham [23, Chapter 16]). It is then easy to compute the value of the option on the discretized asset value process by proceeding backward on the binomial tree. At the expiration date, for $n = 1, \dots, M$, we set $V_M^n = (K - S_M^n)^+$, and then, we recursively compute, for $0 \le n \le i,\ i = 1, \dots, M-1$,

$$V_n^i = \max\left\{\left(K - S_n^i\right)^+, \beta_{\Delta t}\left(pV_{n+1}^{i+1} + (1-p)V_n^{i+1}\right)\right\}. \tag{1.4}$$

(1.4) can be seen as an approximation of (1.3), the continuation value being approximated as the discounted expected value of the two children nodes values. The American option value $V_0(S_0)$ is then approximated by V_0^0.

Several tricks have been proposed to improve the method accuracy. Here we follow the suggestions made by Broadie and Detemple [11] to construct the binomial Black-Scholes with Richardson extrapolation (BBSR) method. At the stage $M-1$, the Black-Scholes formula replaces the usual continuation value in (1.4) by

$$V_{M-1}^i = \max\left\{(K - S_{M-1}^i)^+, K\beta_{\Delta t}\Phi(-d_2) - S_{M-1}^i\Phi(-d_1)\right\}.$$

where $\Phi(\cdot)$ is the standard normal cumulative distribution function, with

$$d_1 = \frac{\ln\left(S_{M-1}^i/K\right) + \Delta t(r + \sigma^2/2)}{\sigma\sqrt{\Delta t}}, \quad d_2 = d_1 - \sigma\sqrt{\Delta t}.$$

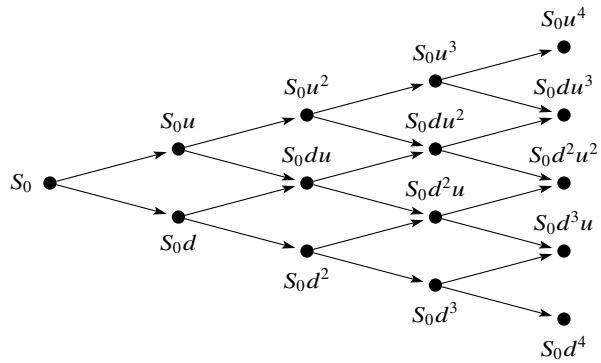

Fig. 1.1: Binomial tree

They furthermore add a two-point Richardson extrapolation consisting in pricing the option with M and $2M$ time steps. Denote by V_M and V_{2M} the respective prices. The approximate option price is then set to $\hat{V}_0 = 2V_{2M} - V_M$.

Efforts have also been made to improve the accuracy by using more complex structures, in particular trinomial trees, but they have not delivered any advantage over binomial trees [14], and therefore are not considered here.

1.4 Dynamic programming approach

The binomial tree method provides a simple and effective method for put options on a single asset following the Black-Scholes model (1.2), and we will use it as a benchmark in our numerical experiments. However, the approach cannot be easily extended to put an option on multiple assets that are not subject to the Black-Scholes model. The other methods that we consider rely on the dynamic programming principle and Monte Carlo simulation [21]. The basic idea consists to sample N price trajectories from the initial asset value S_0, by drawing the asset value S_i^n at time t_i, producing the asset values $\{S_0, S_1^n, \ldots, S_{i-1}^n\}$, for $i = 1, \ldots, M, n = 1, \ldots, N$, and estimate the expected continuation values at each time step for each scenario. Given a scenario and a time stage i, we exercise the option if we obtain a better return value.

We first compute the option value at the expiration date T, for each trajectory $n = 1, \ldots, N$, as

$$\hat{V}_T^n(S_M^n) = \left(K - S_M^n\right)^+ .$$

We then compute estimators of the option value at each time step for each simulated trajectory, using a backwards recursion. For $i = M - 1, \ldots, 1, n = 1, \ldots, N$, we set

$$\hat{V}^n_{t_i}(S_i) = \begin{cases} X_{t_i}(S^n_i) & \text{if } X_{t_i}(S^n_i) \geq \tilde{V}_{t_i}(S^n_i), \\ \beta_{\Delta t_i}\hat{V}^n_{t_{i+1}}(S^n_i) & \text{otherwise,} \end{cases} \tag{1.5}$$

where $\tilde{V}_{t_i}(S_i)$ is an estimation of the continuation value at time t_i, given the current asset value S_i, and

$$\beta_{\Delta t_i} = \frac{\beta_{t_i}}{\beta_{t_{i+1}}}.$$

We finally build the estimator of the option value $V_0(S_0)$ as the maximum between the exercise price at time 0 and the empirical average of the estimated continuation values over the N simulated trajectories, as

$$\hat{V}(S_0) = \max\left\{X_0(S_0), \beta_{t_1}\frac{1}{N}\sum_{n=1}^{N}\hat{V}^n_1\right\}. \tag{1.6}$$

The methods differ in the way we compute $\tilde{V}_{t_i}(S_i)$, $i = 1, \ldots, M-1$ in (1.5).

1.4.1 Regression methods

Longstaff and Schwartz [33] propose to build a function predicting the continuation value by assuming that the latter can be expressed as a linear combination of a countable set of basis functions: for $i = 1, \ldots, M-1$,

$$\beta_{\Delta t}\mathbb{E}\left[V_{t+\Delta t}(S_{i+1}) \mid S_i\right] = \sum_{j=1}^{\infty}\alpha^j_t F^j_t(S_i). \tag{1.7}$$

In order to make it computationally feasible, the sum in (1.7) is truncated to the J first terms, assuming that the truncation error can be neglected:

$$\beta_{\Delta t}\mathbb{E}\left[V_{t+\Delta t}(S_{i+1}) \mid S_i\right] \approx \sum_{j=1}^{J}\alpha^j_t F^j_t(S_i). \tag{1.8}$$

The weights α^j_t are then estimated using a linear regression, i.e. by minimizing the mean square error

$$\min_{\alpha_t}\frac{1}{\#\mathcal{N}_t}\sum_{n\in\mathcal{N}}\left(\sum_{j=1}^{J}\alpha^j_t F^j_t\left(S^n_t\right) - \beta_{\Delta t}\hat{V}^n_{t+\Delta t}\left(S^n_{t+\Delta t}\right)\right)^2, \tag{1.9}$$

where $\mathcal{N}_t$ is the subset of the N simulated trajectories for which the position at time t is in the money, i.e. for any $n \in \mathcal{N}_t$, $K > S^n_t$, while for any $n \in N \setminus \mathcal{N}_t$, $K \leq S^n_t$. This limits the size or the problem to solve, but also reflects that for trajectories out or at the money at time t, it is always preferable to wait rather than to exercise the

option. Various choices can be made for the basis functions [39]. The simplest is to fit a multivariate polynomial in the case of an option over a multidimensional portfolio [21]. For an option over a single asset, we will simply fit a polynomial of degree d.

The continuation value can also be estimated by means of machine learning regression techniques. We first consider the random forest method, a supervised machine learning algorithm widely used in classification and regression problems and initially introduced by Ho [24]. We here focus on the main ingredients, referring to Hastie et al. [22, Chapter 15] for more details. At each time step t_i, $i = 1, \ldots, M-1$, we generate K bootstrap samples, and for each of them, we construct a regression tree [27, Chapter 8], aiming to predict the value of the dependent variable, here the continuation value, as a function of the independent variable, the asset value for each in-the-money trajectory at the time $t_i + \Delta t$. The approach is similar to the expression (1.8), but with a non-parametric model, the weights being replaced by the tree structure. First, a large tree is obtained by means of recursive binary splitting, as each non-terminal node $\mathfrak{n}$ is associated with to a subset of the asset values at time t_i, and is split into two children nodes, gathering the values either less or greater than some value $s_{\mathfrak{n}}$, typically the mean of the values associated with to the father node. We stop when each terminal node has less than some minimum number of observations. The tree size is then reduced by applying cost complexity pruning, in order to limit the number of leaf nodes, knowing that more nodes tend to provide a better estimation, but a greater risk of overfitting. Each terminal node then provides a prediction of the dependent variable value, i.e. the continuation value, and the predictions are averaged to produce the output of the tree. We can then use this built collection of trees to compute a set of predictions $\mathfrak{v}_{t_i}^k(S)$, $k = 1, \ldots, K$, given the asset value S, and average them to produce the estimation of the continuation value as

$$\tilde{V}_{t_i}(S) = \frac{1}{K} \sum_{k=1}^{K} \mathfrak{v}_{t_i}^k(S).$$

Another class of models popular in machine learning are neural networks. A neural network is a directed graph where each node represents a neuron, that receives a number of inputs x_i from its direct predecessors and applies an activation function to produce an output that feeds its direct successors. Several activation functions exist. We here choose the ReLU function, defined as $ReLU(x) = x^+$, as it guarantees that only non-negative values are produced. We also limit ourselves to multilayer perceptions (MLP) [22, Chapter 11], where each neuron receives the output of the previous layer as input, and its own output is the input of the next layer, as illustrated in Figure 1.2.

At each stage t_i, $i = 1, \ldots, M-1$, we train a neural network by minimizing a loss function, here defined similarly to (1.9) as the mean square error minimization problem

$$\min_{\gamma_{t_i}} \frac{1}{\#\mathcal{N}_{t_i}} \sum_{n \in \mathcal{N}_{t_i}} \left(\tilde{V}_{t_i}\left(\gamma_{t_i}, S_{t_i}^n\right) - \beta_{\Delta t_i} \hat{V}_{t_i+\Delta t_i}^n \left(S_{t_i+\Delta t_i}^n\right)\right)^2, \tag{1.10}$$

for the $\#\mathcal{N}_{t_i}$ in-the-money scenarios at stage t_i. We then set

$$\tilde{V}_{t_i}(\cdot) = \tilde{V}_{t_i}(\gamma^*_{t_i}, \cdot),$$

where $\gamma^*_{t_i}$ is the optimal solution found when solving (1.10).

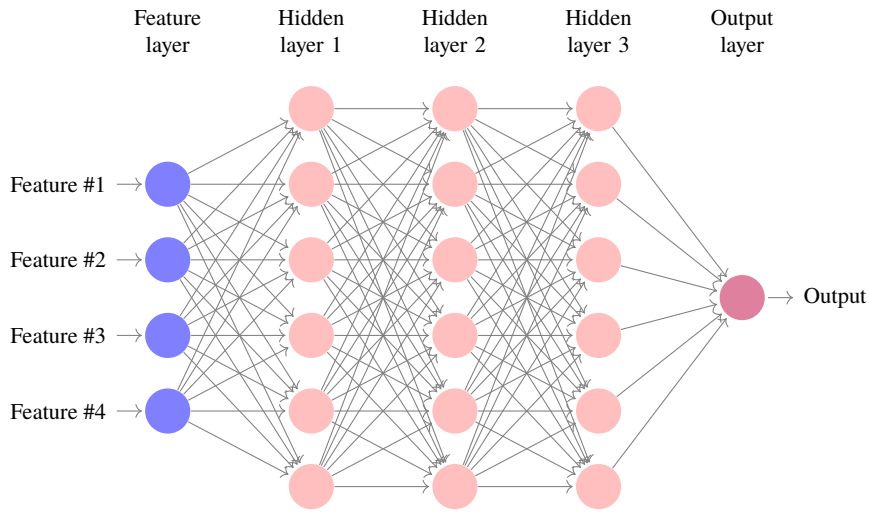

Fig. 1.2: The architecture of a neural network

The machine learning architectures have to be trained on a large number of scenarios (we consider 100000 scenarios in our experiments), leading to a significant overhead time. Once trained, they can be used on several replications of the option pricing problem, as long as the underlying asset value follows the same process.

1.4.2 Malliavin calculus

Fournié et al. [19] express the expected value of the option at time t, conditionally to the asset value at time $s \leq t$, as

$$\mathbb{E}\left[V_t(S_t) \mid S_s = \alpha\right] = \frac{\mathbb{E}\left[\pi_{s,t}(V_t(S_t), \alpha)\right]}{\mathbb{E}[\pi_{s,t}(1, \alpha)]} \tag{1.11}$$

where $\pi_{s,t}(\cdot)$ are the Malliavin weights, whose expressions vary according to the asset value process. Under the Black-Scholes model (1.2), Bally et al. [2] show that

$$\pi_{s,t}(\xi, \alpha) = \frac{\xi H(S_s - \alpha)}{\sigma s(t-s) S_s} \Delta B_{s,t},$$

where $\Delta B_{s,t} = tB_s - sB_t + \sigma s(t-s)$, and $H(\cdot)$ is the Heaviside step function, defined as $H(\ell) = \mathbb{1}(\ell \geq 0)$, $\mathbb{1}$ representing the indicator function. They further refine the computation of the conditional expectation (1.11) by introducing a density function ψ parameterized by $x \geq 0$, referred as the localization function, leading to

$$\mathbb{E}[V_t(S_t) \mid S_s = \alpha] = \frac{\mathbb{T}^{\psi}_{s,t}(V_t(S_t), \alpha)}{\mathbb{T}^{\psi}_{s,t}(1, \alpha)} \tag{1.12}$$

with

$$\mathbb{T}^{\psi}_{s,t}(\xi, \alpha) = \mathbb{E}[\xi\psi(S_s - \alpha)] + \mathbb{E}\left[\xi \frac{H(S_s - \alpha) - \Psi(\xi, S_s - \alpha)}{\sigma s(t-s) S_s} \Delta B_{s,t}\right],$$

where $\Psi(x, \ell) = \int_{-\infty}^{\ell} \psi(x, y) dy$ is the cumulative distribution function associated to $\psi(x, \cdot)$. We follow their suggestion and consider in our numerical experimentation the Laplace-type probability distribution function

$$\psi(x, y) = \frac{\lambda(x)}{2} e^{-\lambda(x)\ |y|} \tag{1.13}$$

where $\lambda(x) = 1/\sqrt{t-s}$.

Bally et al. [3] adapt the algorithm proposed by Lions and Régnier [32] and using a backwards recursion, they approximate the N continuation values $\mathbb{E}[V_{t_{i+1}}(S_{i+1}) \mid S_i = S_i^n]$, $n = 1, \ldots, N$, at time t_i, by taking the empirical sample averages of the expectations appearing in the right hand sides of (1.11) and (1.12) over the N simulated asset price trajectories. Ignoring the localization function for simplicity of exposition, the estimated discounted continuation value is

$$\tilde{V}_{t_i}(S_i^n) = \frac{\hat{\pi}_{t_i, t_{i+1}}(\hat{V}_{t_{i+1}}, S_i^n)}{\hat{\pi}_{t_i, t_{i+1}}(1, S_i^n)},$$

with

$$\hat{\pi}_{t_i, t_{i+1}}(\hat{V}_{t_{i+1}}, S_i^n) = \frac{1}{N} \sum_{q=1}^{N} \frac{\hat{V}^q_{t_{i+1}}(S^q_{i+1}) H\left(S_i^q - S_i^n\right)}{\sigma t_i \Delta t S_i^q} \Delta B^q_{t_i, t_{i+1}},$$

$$\hat{\pi}_{t_i, t_{i+1}}(1, S_i^n) = \frac{1}{N} \sum_{q=1}^{N} \frac{H\left(S_i^q - S_i^n\right)}{\sigma t_i \Delta t S_i^q} \Delta B^q_{t_i, t_{i+1}},$$

where $\Delta B^q_{t_i, t_{i+1}}$ is obtained from the draws generating the qth asset value path. The computational effort required at each time step is therefore of order $O(N^2)$.

1.5 Control variates

The convergence properties of the Longstaff and Schwartz's algorithm have been studied in details [15]. In essence, when N and M grow to infinity, (1.6) converges to a lower bound of the American option value. The quality of this bound can be refined and made arbitrarily close to the true value by improving the accuracy of the approximations $\tilde{V}_t(\cdot)$, $t \in cT$, especially by letting J rise to infinity in (1.8). However, for finite N, M, and J, the error can be large and the estimator (1.6) can present a significant noise. If M is too small, the number of opportunities to exercise the option is limited and we tend to underestimate the option value, while if N is not large enough, we face the risk of overfit the policy with respect to the simulated trajectories, and to overestimate the option value. The quality of the estimator (1.6) can be improved by applying variance reduction techniques [6, 30], among which control variates have received a lot of attention for option pricing. Given an estimator X, the basic idea is to find an random variable Y whose expectation is known, and highly correlated with X. We then form the new estimator $Z = X - \theta(Y - E[Y])$. If θ is fixed, $E[Z] = E[X]$ and $\mathrm{Var}[Z] = \mathrm{Var}[X] + \mathrm{Var}[Y] - 2\theta\,\mathrm{Cov}(X,Y)$. The variance of Z can be minimized by choosing

$$\theta = \frac{\mathrm{Cov}(X,Y)}{\mathrm{Var}[Y]}.$$

θ can be estimated using pilot experiments, but more often, an estimator will be produced using the same draws as those used to generate X and Y, so that Z is biased, but usually, this bias is negligible. When θ is set to one, we say that Z is an indirect estimator, noticing that several authors still speak about control variate [25, 3, 13].

It is often possible to accurately estimate the price of an European option. When the asset value follows a simple process, as in the Black-Scholes model, it is even possible to derive it analytically. This suggests to use the European price estimator $\hat{V}_0^E = \beta_T (K - S_T)^+$ as a control variate for the American option price estimator [25]. Its performance is however disappointing, and several authors have searched for better candidates [8, 18]. Rasmussen [39] notices that an early exercise produces a value whose correlation with the corresponding European option price degrades as the expiration date is later in the future. He then suggested to use as a control variate for the continuation value an European option emitted at the candidate exercise time of the American option. Indeed, the optional sampling theorem establishes that for any scenario n, the exercise value at T is an unbiased estimator of the price of the European option emitted at the exercise time τ_n of the American option under this scenario, and at time 0 the American option is never exercised.

In order to apply this control variate, we first initialize the estimator of the European option at the expiration date T as $V_T^{E,n} = X_T^n(S_T^n)$. Going backwards, for $i = M-1, \ldots, 1$, in addition to $\tilde{V}_{t_i}(S_i^n)$, we produce an estimator $\tilde{V}_{t_i}^E(S_i^n)$ of the value of the European option starting from t_i and expiring at T, for $n = 1, \ldots, N$, using the same estimation technique. We also estimate the second moment of this estimator and its covariance with the continuation value estimator, in order to get $\theta_{t_i}^n$. We then

replace $\tilde{V}_{t_i}(S_i^n)$ by $\tilde{V}_{t_i}(S_i^n) - \theta_{t_i}^n(\tilde{V}_{t_i}^E(S_i^n) - V_{t_i}^E(S_i^n))$ to estimate the continuation value at time t_i for the scenario n. At time 0, the European option theoretical price is computed for each scenario, using the exercise time of the American option as the emission date of the European option, and a time 0 if is never exercised, leading to the new American option price estimator

$$\hat{Z}_0 = \max\left\{X_0(S_0), \frac{1}{N}\sum_{n=1}^{N}\left(\beta_{t_1}\hat{V}_1^n - \theta_0\left(\beta_{\tau_n}V_{\tau_n}^{E,n}(S_{\tau_n}^n) - V_0^E(S_0)\right)\right)\right\},$$

where τ_n is set to T if the option is never exercised in the scenario n. For more implementation details, we refer the reader to West [42, Chapter 3].

1.6 Numerical experiments

We now evaluate the presented methods to estimate the price of American options. Similar results have been obtained for the Delta, but are not reported as they bring similar conclusions. As a convention, we abbreviate the method name by LS for the Longstaff and Schwartz's algorithm, RF for the Random Forests, NN for the Neural Networks, and Malliavin for the estimation based on Malliavin calculus. The experiments have been conducted using Python 3.9 and numpy and sklearn libraries, on an Intel i7-9700K at 3.60GHz, eight cores, and 32Go of memory. Random numbers have produced using the xoshiro pseudo-random generator [7]. Similar results have been obtained with the MRG32k3a generator [29]. We followed Bally et al. [2] when implementing the Malliavin calculus approach, using the localization function (1.13) and a pointwise indirect estimation for the computation of the continuation value, based on the European option value emitted at the exercise time. Results obtained with indirect estimation are identified with the postfix "-I", and with the suffix "-CV" for those produced with the Rasmussen's control variates. The impementation code is available at `https://github.com/RaulChavezAquino/Monte-Carlo-methods-for-pricing-American-Options`.

We first consider an American put option on the stock price process S with parameters $S_0 = 100$, $K = 100$, $\sigma = 0.20$, $r = \ln(1.1)$, $T = 1$, taken from Bally et al. [3]. The binomial tree gives us a reference option value of 4.9175.

We report the estimation time as a function of M and N for each technique in Figure 1.3. The MLP models have been trained using sklearn and the solver ADAM. For each time discretization factor $M = (10, 20, 50, 100, 500)$, we use $(2, 2, 2, 3, 3, 3)$ hidden layers and $(10, 20, 50, 25, 50, 70)$ neurons per layer, respectively, those configurations having been selected by trial and error. The random forests have been trained with 10 trees and a maximum number of leaves equal to 15. The machine learning models have been trained by 100000 scenarios, but the training time is not taken into account in these graphs, while it affects the overall performance. For the random forests, we observed an overhead due to the training going from 10 seconds for $(M, N) = (10, 500)$ to 235 seconds for $(M, N) = (200, 10000)$. We have also

empirically observed that this overhead approximately grows linearly with M, but does not significantly change with N. MLP models required a large training time, and we stopped the computations when it exceeded 2 hours. As a result, we only report results for $M = 10$ and $M = 20$ with the neural networks. From Figure 1.3, we observe that LS is the fastest method while Malliavin calculus requires prohibitive times. The effect could be exacerbated due to the use of Python, an interpreted programming language, but we observe that the computation time rises at a rate faster than a linear rate with the number of scenarios, as expected, while the other approaches exhibit an computation time that grows approximately linearly with N.

We next analyze in more details the performances of the methods when we vary N and M, for the naive estimator, the indirect estimator, and the estimator with the control variate. For each configuration, we repeat the valuation 1000 times, and draw a box plot over these replications. Figures 1.4 and 1.5 show the behavior of LS and RF methods, respectively, where we fix $N = 5000$ on the left part and $M = 100$ on the right part. When M is small and N is kept constant, we tend to underestimate the option value, but we can face some overfit when M is large with LS. On the other side, fixing M reveals a large overfit for small values of N with LS, but a convergence to a lower bound of the option price when N grows. Several configurations were tried with the random forests. The two upper graphs in Figure 1.5 have been obtained with 50 trees and a maximum number of leaves of 20, while the two lower graphs have been computed with 90 trees and a maximum of 100 leaves. We see that the random forests always underestimate the true price, and the effect can be dramatic with the naive estimator when we have many leaves. However, applying variance reduction techniques allow to obtain good price estimators, and the estimators obtained with 100 leaves are close to the ones obtained with the least-squares approach. Large random forests therefore appear very sensitive to the variance in the asset price trajectories, and more research is needed to properly explain the method behavior, and to automatically choose the best design. Reducing the variance nevertheless allows the random forests to be competitive, and they represent an alternative method to parametric regression that we plan to further investigate. Of course, the option value can also be computed during the training phase of the model, and for large M and N, random forests deliver good price estimation, even with a simple configuration of 10 trees and a maximum of 15 nodes used to produce the results reported in Table 1.1, computed over one run, where we additionally report in brackets the standard deviation over the N simulated trajectories.

We finally illustrate the MLP model with $N = 1000$, $M \in \{10, 20, 50, 100, 200\}$, and the Malliavin calculus, with $M = 100$, and N going from $N = 100$ to 20000, in Figure 1.6. In both cases, the computational burden prevented us to perform more experiments. We first observe that the MLP model strongly underestimates the option value, the control variate performing better than the naive estimator, but less than the indirect estimation, that is still far from the quality obtained with the LS and RF approaches. Better MLP designs could improve the results, but we do not have an automatic way to find a good MLP structure. On the other hand, the Malliavin calculus gives good price estimation when the number of scenarios is at least equal

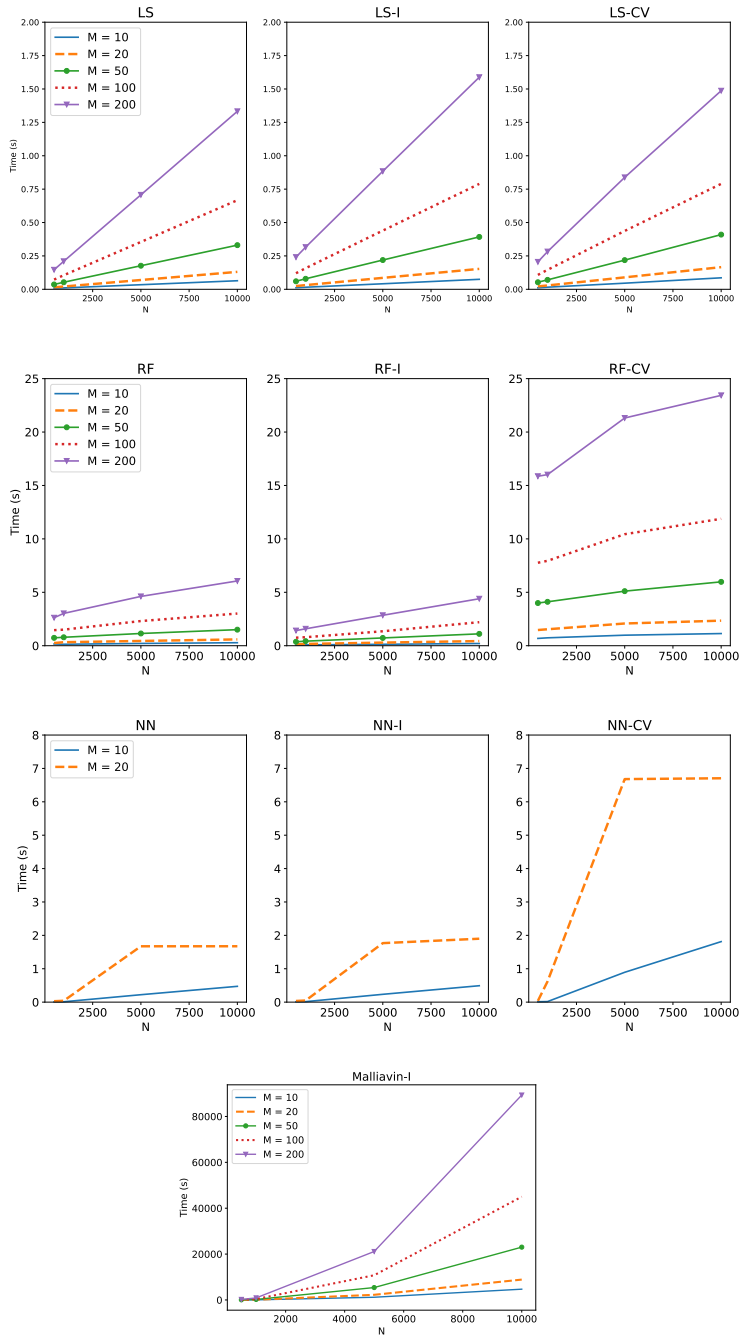

Fig. 1.3: Estimation time (s)

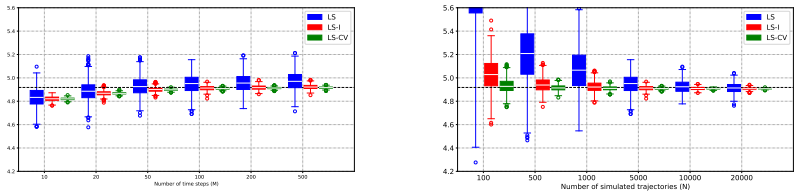

Fig. 1.4: Least-squares regression

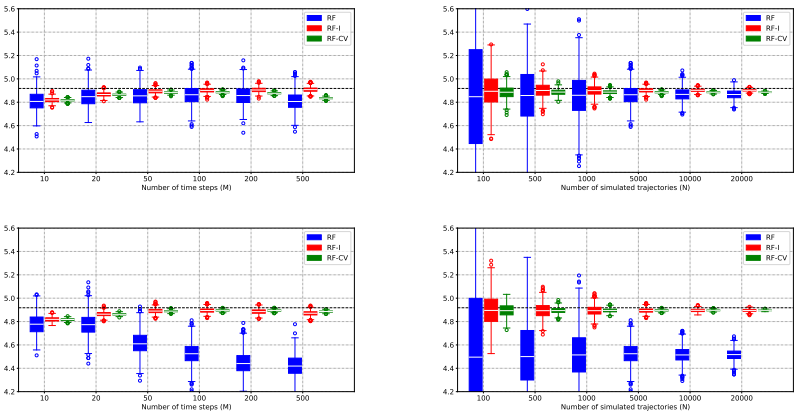

Fig. 1.5: Random forests

M	N	RF	RF-I	RF-CV
10	10000	4.89 (6.01)	4.82 (1.36)	4.82 (0.65)
	100000	4.71 (5.99)	4.82 (1.36)	4.82 (0.64)
20	10000	4.90 (5.99)	4.87 (1.38)	4.86 (0.60)
	100000	4.77 (5.91)	4.86 (1.38)	4.86 (0.61)
50	10000	4.98 (5.97)	4.92 (1.40)	4.89 (0.58)
	100000	4.85 (5.89)	4.91 (1.41)	4.89 (0.58)
200	10000	5.04 (5.81)	4.92 (1.38)	4.90 (0.56)
	100000	4.89 (5.68)	4.92 (1.37)	4.90 (0.55)

Table 1.1: American option value with RF

to 5000. However, the LS and RF models already perform well with $N = 1000$, and therefore, on this example, the Malliavin calculus does not provide any advantage.

We finally report the estimated American option values on five real stocks, whose characteristics are reported in Table 1.2. The data were obtained from Quandl, now replaced by Data Nasdaq Link (`https://data.nasdaq.com/`). We first compute the option value for $N \in \{500, 1000, 5000, 10000, 20000\}$ and $M \in \{10, 20, 50, 100, 200, 500\}$, using the LS, RF, and NN approaches. We do not

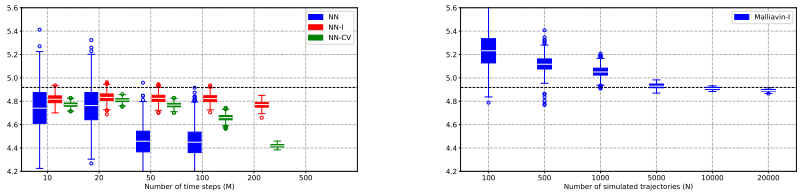

Fig. 1.6: Neural networks (left) and Malliavin-I (right)

report results with the Malliavin calculus due to the required computational time to produce them. From Figure 1.7, showing all the options except GOOG due to scale difference, we see that pure LS method delivers better results than RF, which itself performs better than NN. However, all the methods provides similar results, close to the true option price, as soon as indirect estimation or control variates are used, even for small values of M and N. The time required to train the neural networks prevented us to report the MISFT value for N greater than 10000 and M greater than 100. In order to better assess the effect of variance reduction, we finally report in Table 1.3 the option values averaged over 5000 simulated paths, along with the standard deviations in brackets, and $M = 100$. Indirect estimation and control variates again dramatically reduce the standard deviations.

Stock	V_0	S_0	K	T	r	σ	Trade Date	Maturity	Name
AAPL	8.65	113.700	100.0	1.17808	0.0086	0.310016	2015-11-17	2017-01-20	Apple
BIIB	13.00	279.095	200.0	1.17808	0.0086	0.412166	2015-11-17	2017-01-20	Biogen
GOOG	135.70	726.160	860.0	0.06575	0.0007	0.410721	2015-11-17	2015-12-11	Google
MSFT	17.75	53.020	80.0	0.41096	0.0034	0.424071	2015-11-17	2016-04-15	Microsoft
SBUX	10.25	60.620	70.5	0.06575	0.0007	0.462461	2015-11-17	2015-12-11	Starbucks

Table 1.2: Real data characteristics

Stock	LS	LS-I	LS-CV	RF	RF-I	RF-CV
APPL	8.14 (11.69)	8.07 (0.08)	8.07 (0.04)	7.88 (11.47)	8.07 (0.08)	8.06 (0.03)
BIIB	13.53 (26.15)	13.23 (0.11)	13.23 (0.05)	13.6301 (24.87)	13.23 (0.11)	13.21 (0.04)
GOOG	136.33 (48.00)	135.72 (0.01)	135.72 (0.01)	134.02 (16.84)	135.72 (0.01)	135.71 (0.01)
MSFT	27.61 (8.91)	27.41 (0.02)	27.41 (0.01)	27.09 (4.36)	27.41 (0.02)	27.39 (0.01)
SBUX	10.46 (5.90)	10.25 (0.00)	10.25 (0.00)	10.01 (2.34)	10.25 (0.00)	10.25 (0.00)

Table 1.3: Estimated option prices, $N = 5000$, $M = 100$.

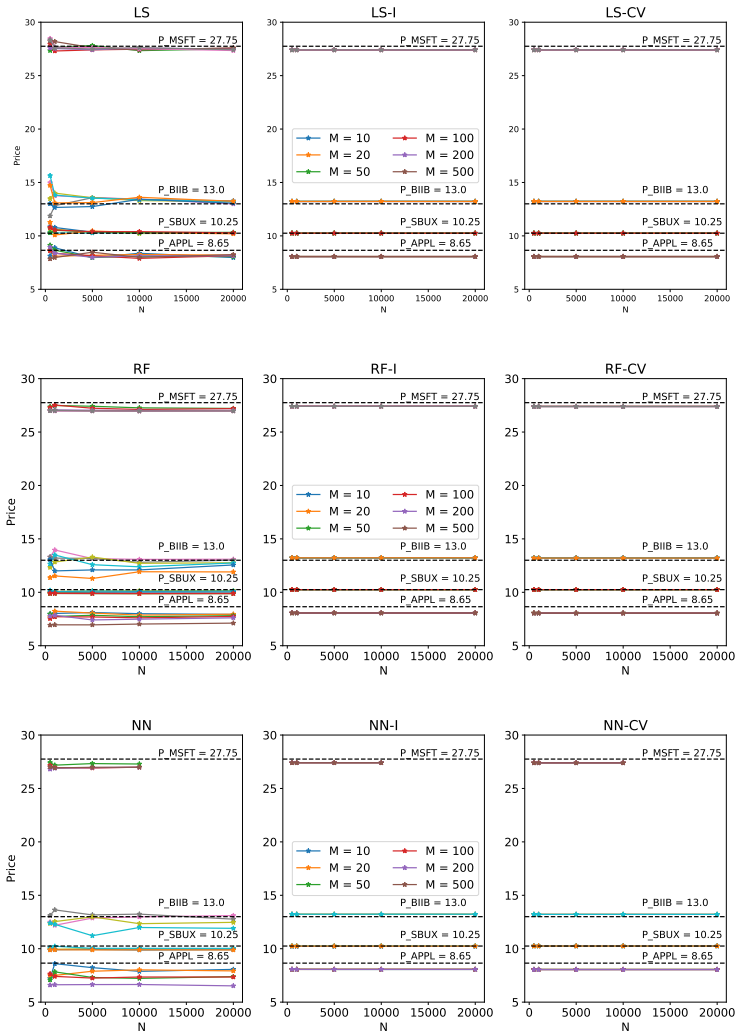

Fig. 1.7: Real option data comparison

1.7 Conclusion

American options valuation has been an active research topic for more than three decades, and many approaches have been proposed to tackle it. We have reviewed and compared a few of them, from the least-squares Monte Carlo method [33] to Malliavin calculus and machine learning techniques, on simple examples. While providing accurate results, Malliavin calculus exhibited a large computational bur-

den in addition to the theoretical difficulties to derive the weights, and for practical purposes, it is outperformed by traditional regression techniques. Advanced machine learning techniques, such as neural networks, require careful designs and more research is needed to validate and automatically select them. In particular, we were not able to obtain satisfying results with multilayer perceptions, while random forests deserve more investigation as well selected structures brought promising results. In both cases, we must face higher calibration costs, requiring a large number of scenarios, but option value can also be estimated during this phase, and the computational effort remains reasonable for random forests. Least-squares Monte Carlo approaches performed remarkably well, but we must keep in mind that we reported results on simple examples only. Variance reduction techniques, illustrated by the control variates proposed by Rasmussen [39], appeared to be a crucial ingredient in any of the investigated methods, dramatically improving the results, at a negligible cost.

We nevertheless must remain cautious as more experiments should be performed on more complex situations, including high-dimensional options and value prices with stochastic volatility of jumps. While the Greeks can be estimated by finite difference, specific techniques have been proposed and could also be considered [26]. Some authors [4] suggest that deep learning approaches help to avoid the curse of dimension. However, it is often difficult to collect enough data to accurately calibrate such models and simulation, along with simplifying assumptions on the assets prices processes, remains a key tool. Carefully designed variance reduction techniques allow the estimation of the option price with a limited number of simulated assets price trajectories, and classical parametric regression techniques appear more robust than complex machine learning models that often require a large number of simulations to calibrate, and can underperform if their design is not well selected. We have focused on control variates, but other variance reduction methods, as scenario bundles [26], multilevel Monte Carlo [5], or quasi-Monte Carlo sampling [17], should be also considered, especially as such techniques can be combined [31].

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